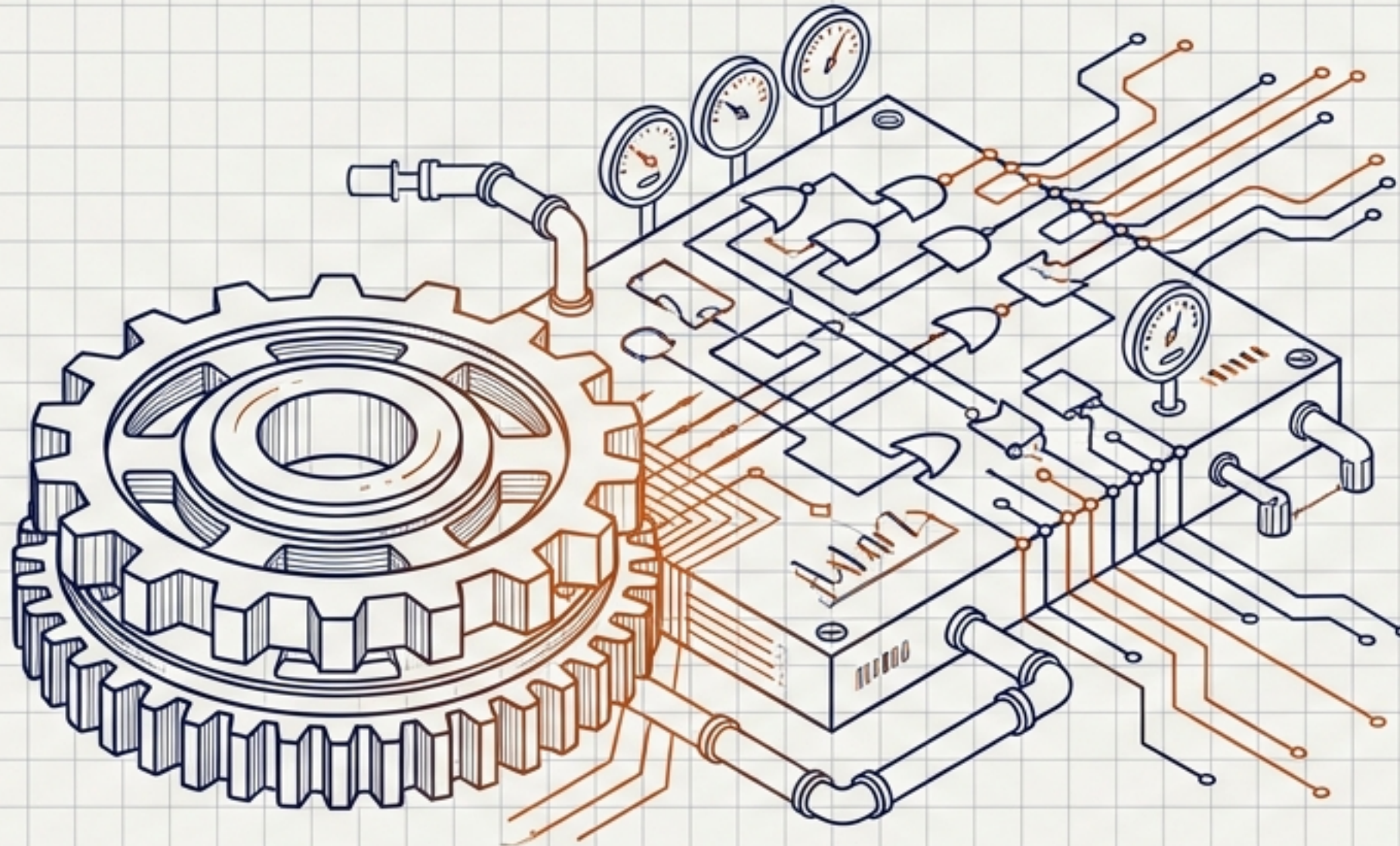


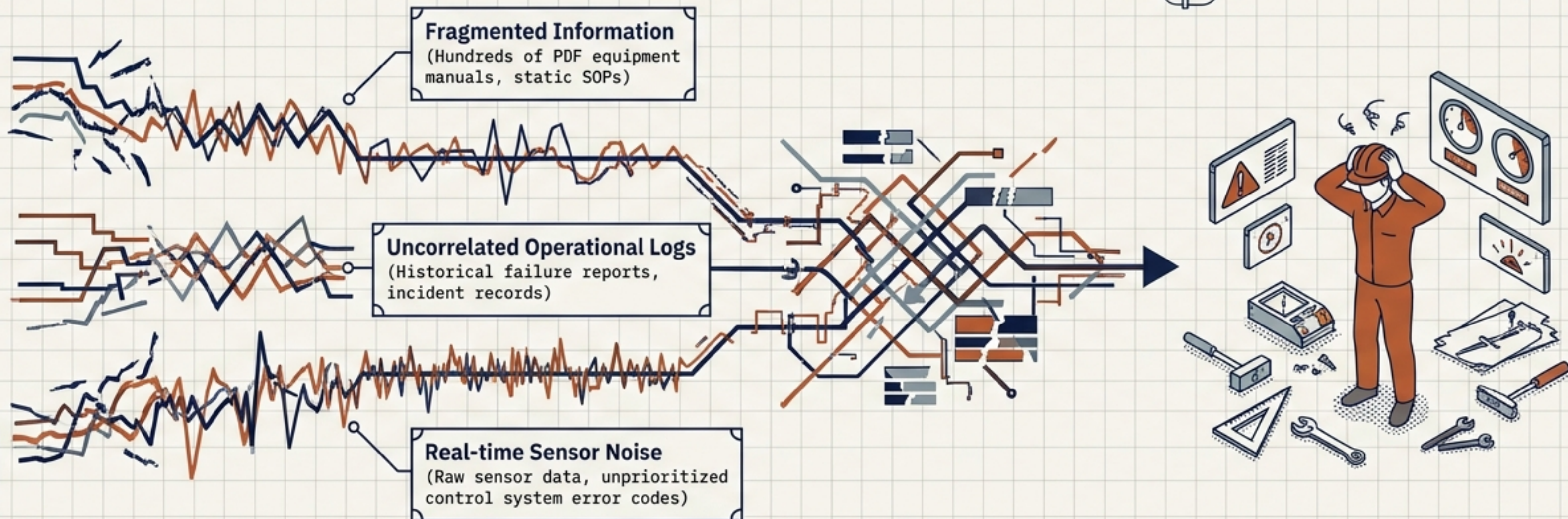


The Intelligent Maintenance Wizard

Engineering the Digital Forge: Architecture & Requirements Blueprint

Hackathon Technical Reference Document

The Current Operational Bottleneck



Unplanned downtime in complex, capital-intensive steel manufacturing results in severe production loss and safety risks. Manual, expert-dependent troubleshooting is too slow and inconsistent for modern industrial demands.

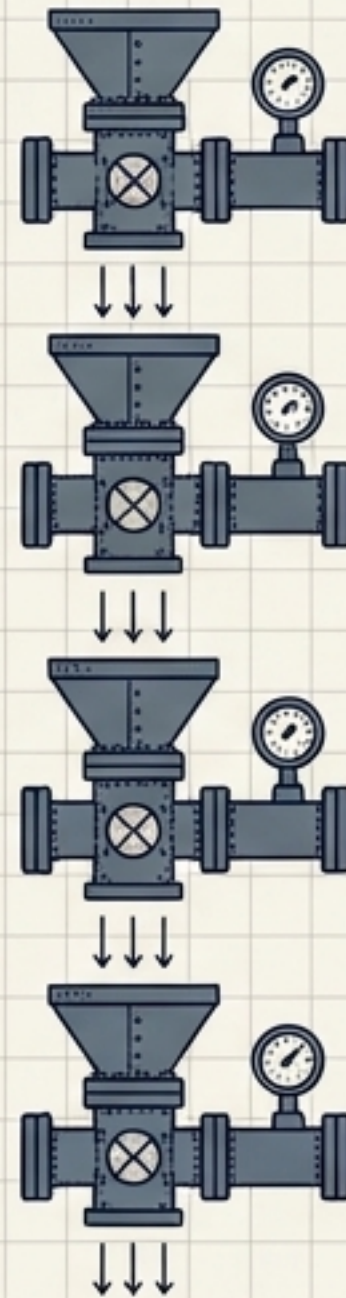
The Paradigm Shift Matrix

	 Current State		Target State
Trigger	Breakdown & Failure	➔	Dynamic Anomaly Detection
Data Source	Siloed & Fragmented PDFs/Logs	➔	Context-Aware, Unified Knowledge Base
Action Generation	Manual Troubleshooting & Expert Judgment	➔	AI-Recommended, Prioritized Steps
Resulting Impact	Reactive Escalation & Downtime	➔	Proactive Intervention & Uptime

The goal is a decision-support platform that shifts the entire industrial operations team from reactive survival to proactive mastery.

Systems Architecture Blueprint

Ingestion Nodes



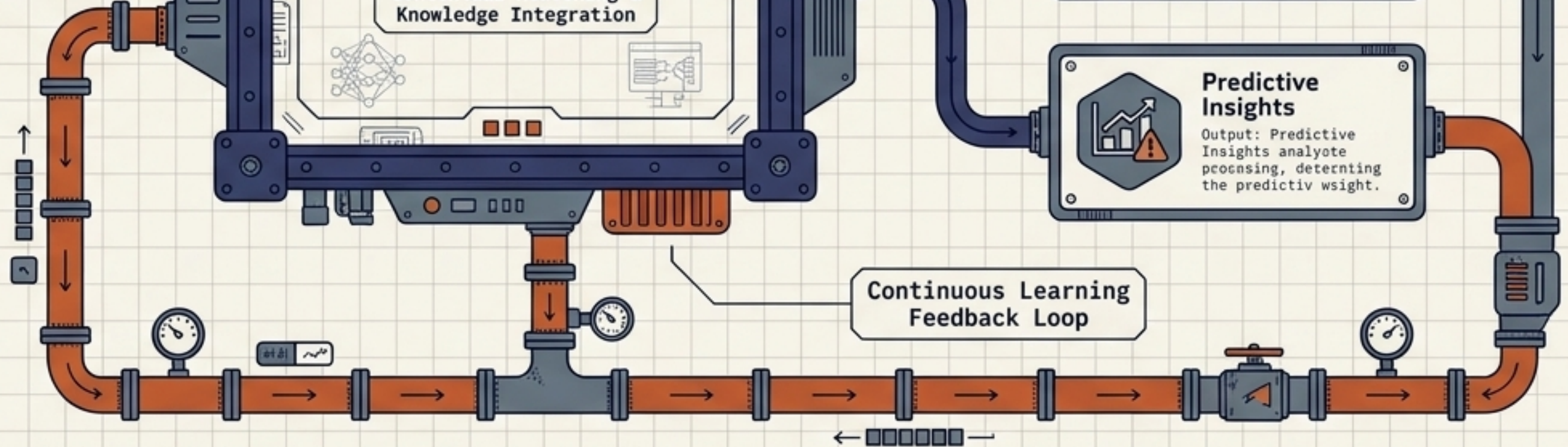
Cognitive Core: LLM/SLM Reasoning Engine

Contextual Reasoning &
Knowledge Integration

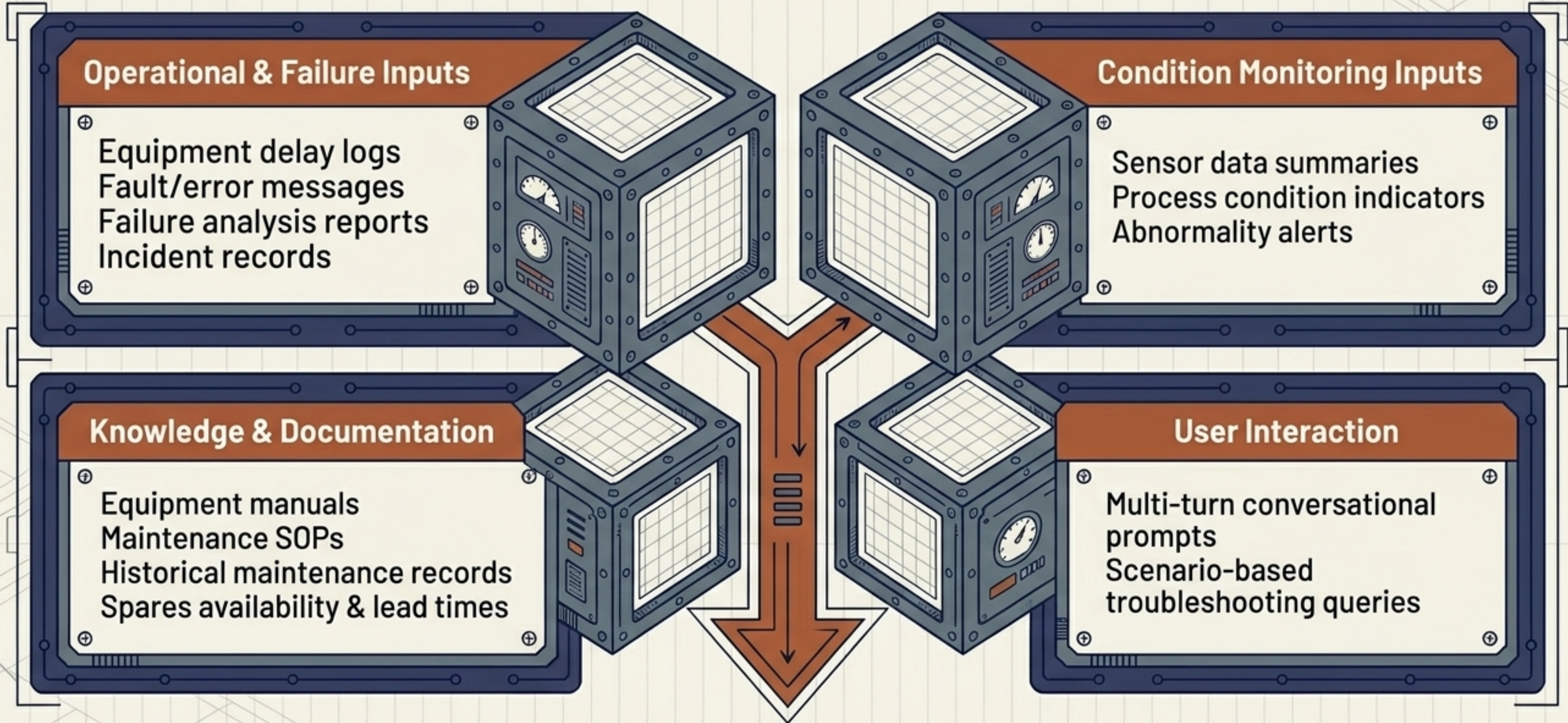
Output Nodes: Explainable Recommendations



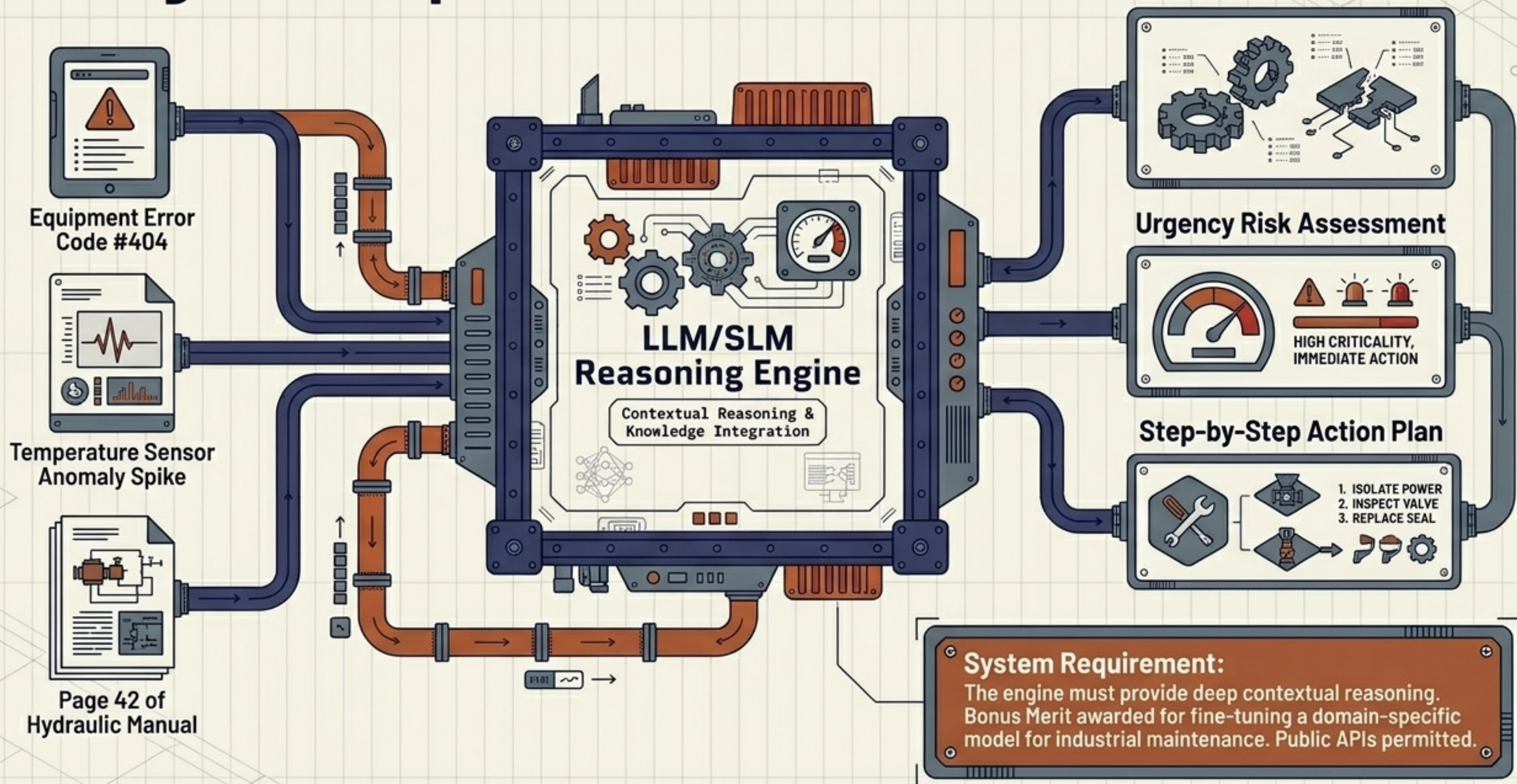
Continuous Learning
Feedback Loop



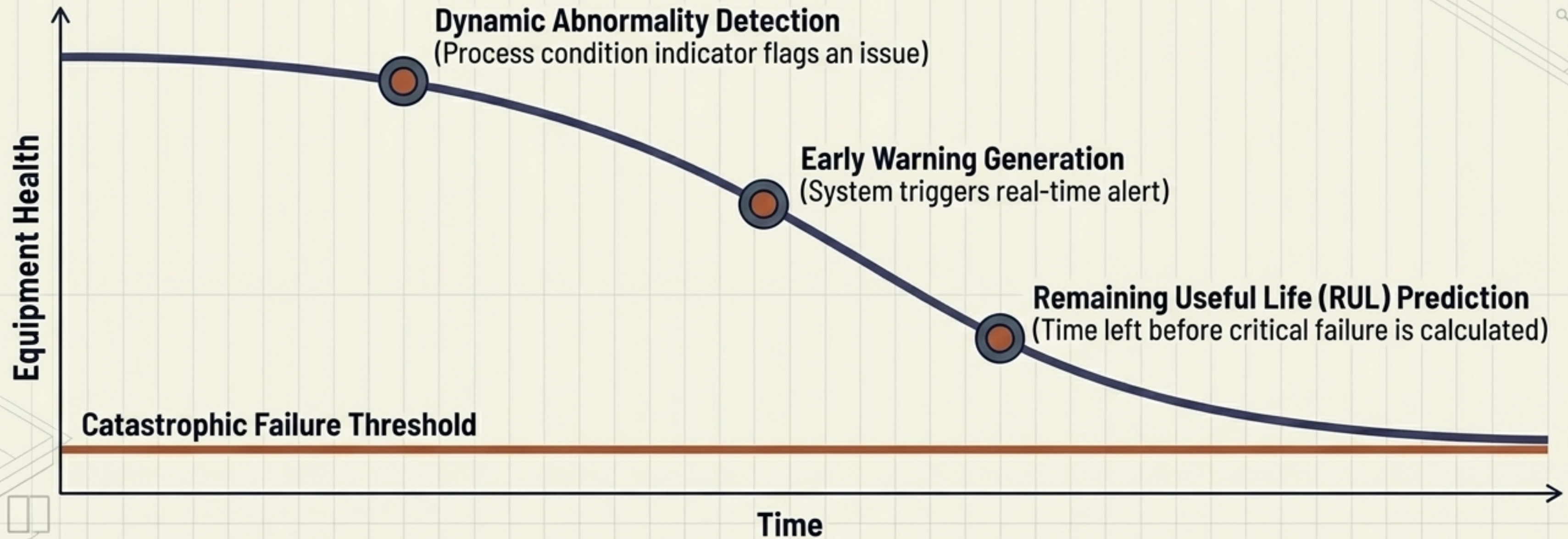
Multimodal Data Ingestion Requirements



The Cognitive Pipeline

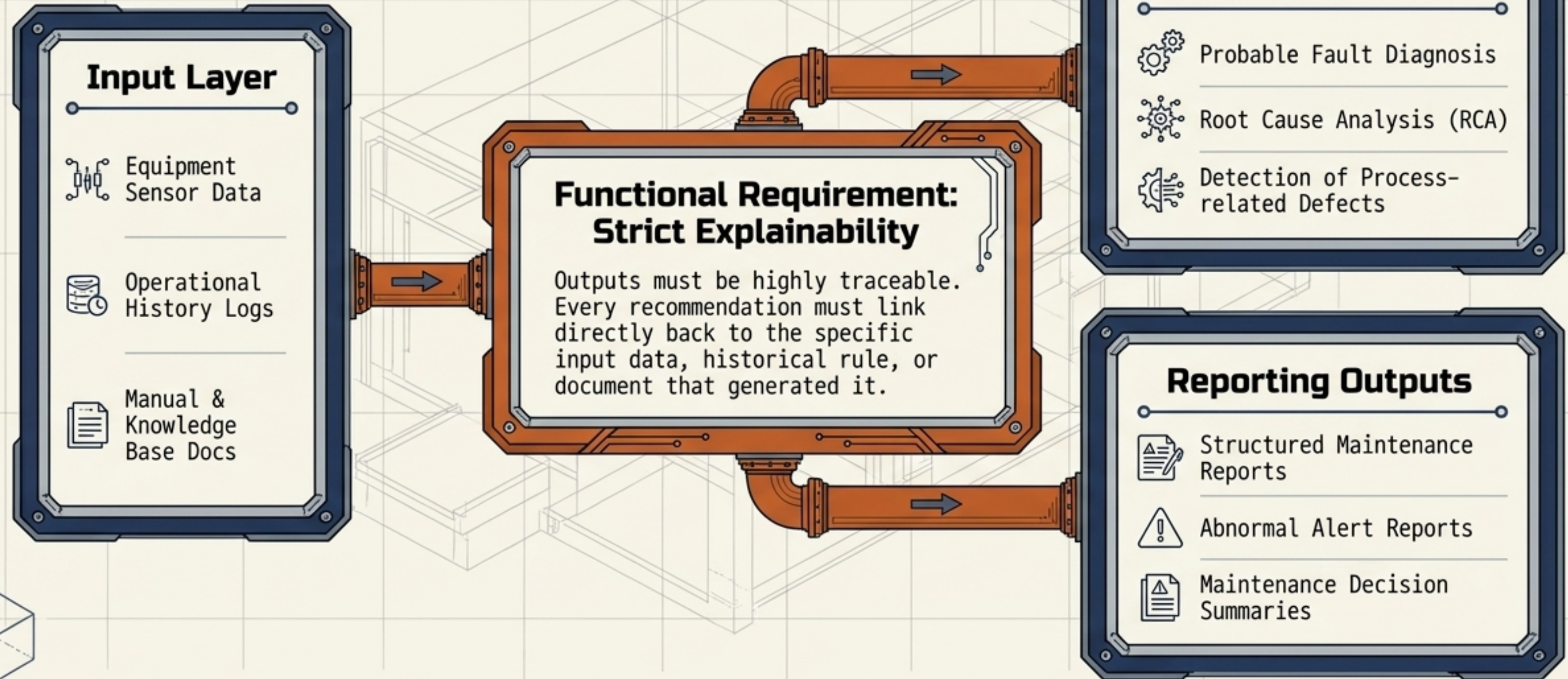


Failure Prediction & Remaining Useful Life (RUL)

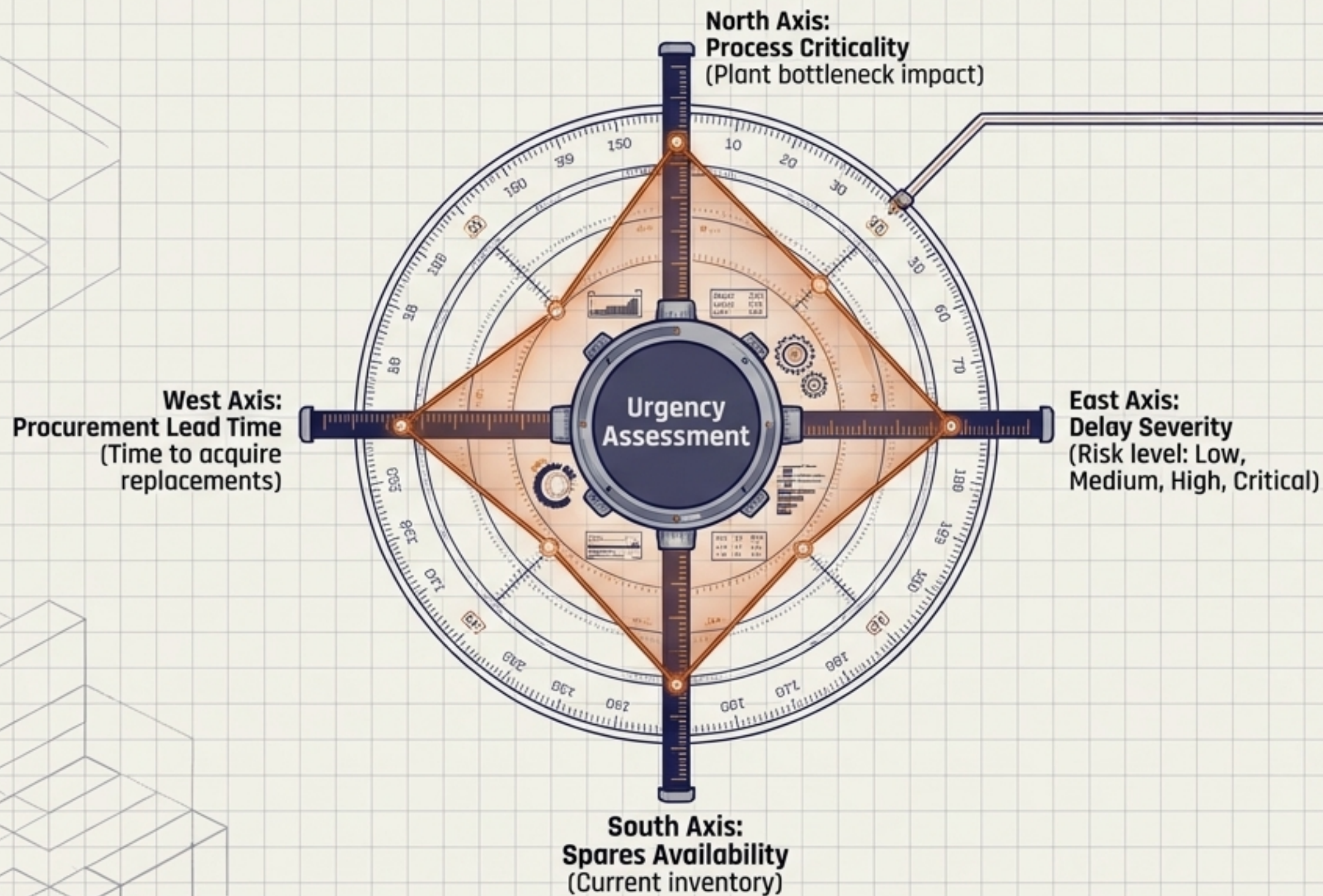


The system must actively monitor health indicators to predict process-related defects before they manifest as critical breakdowns.

Structured Diagnostic & Reporting Modalities



Multi-Dimensional Risk & Priority Engine



Contextual Constraint:

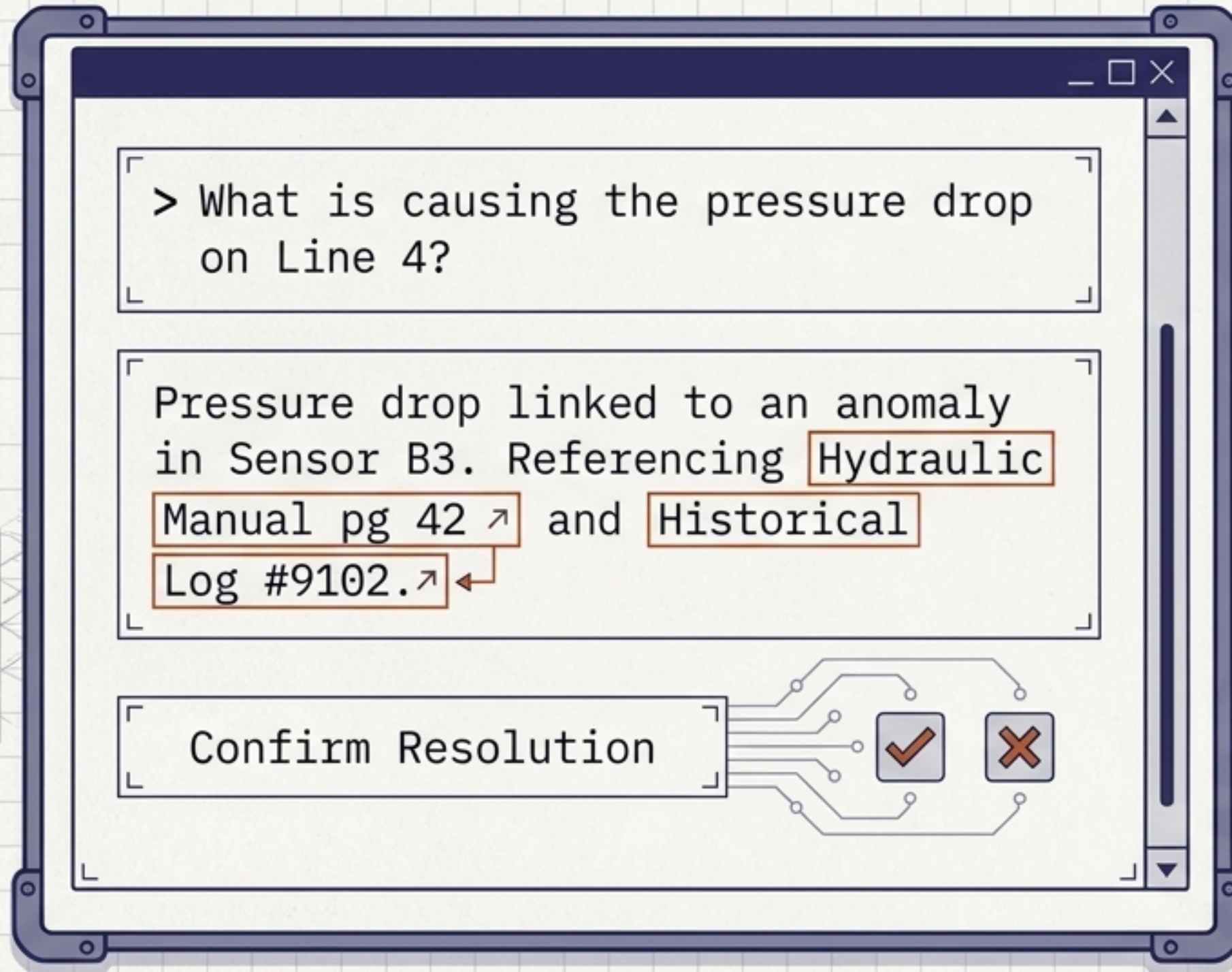
Maintenance actions cannot be recommended in a vacuum. The system must filter and sequence interventions based on real-world operational impact and strict procurement constraints.

Actionable Maintenance Plans



The Wizard must provide structured, sequenced deliverables to the engineer, completely eliminating guesswork on the plant floor.

Natural Language Interaction & Feedback



> What is causing the pressure drop on Line 4?

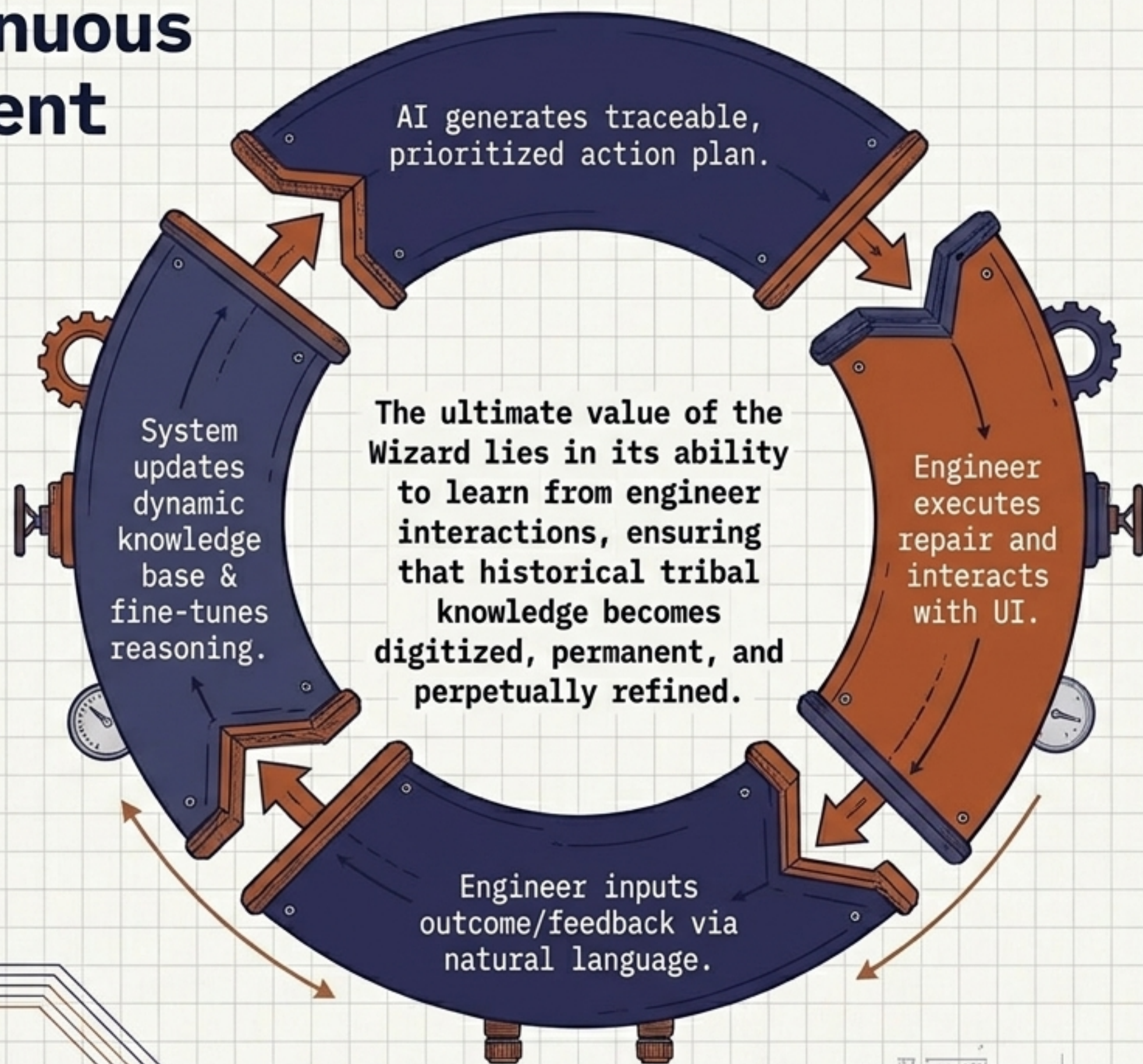
Pressure drop linked to an anomaly in Sensor B3. Referencing **Hydraulic Manual pg 42 ↗** and **Historical Log #9102. ↖**

Confirm Resolution ☒ ☐

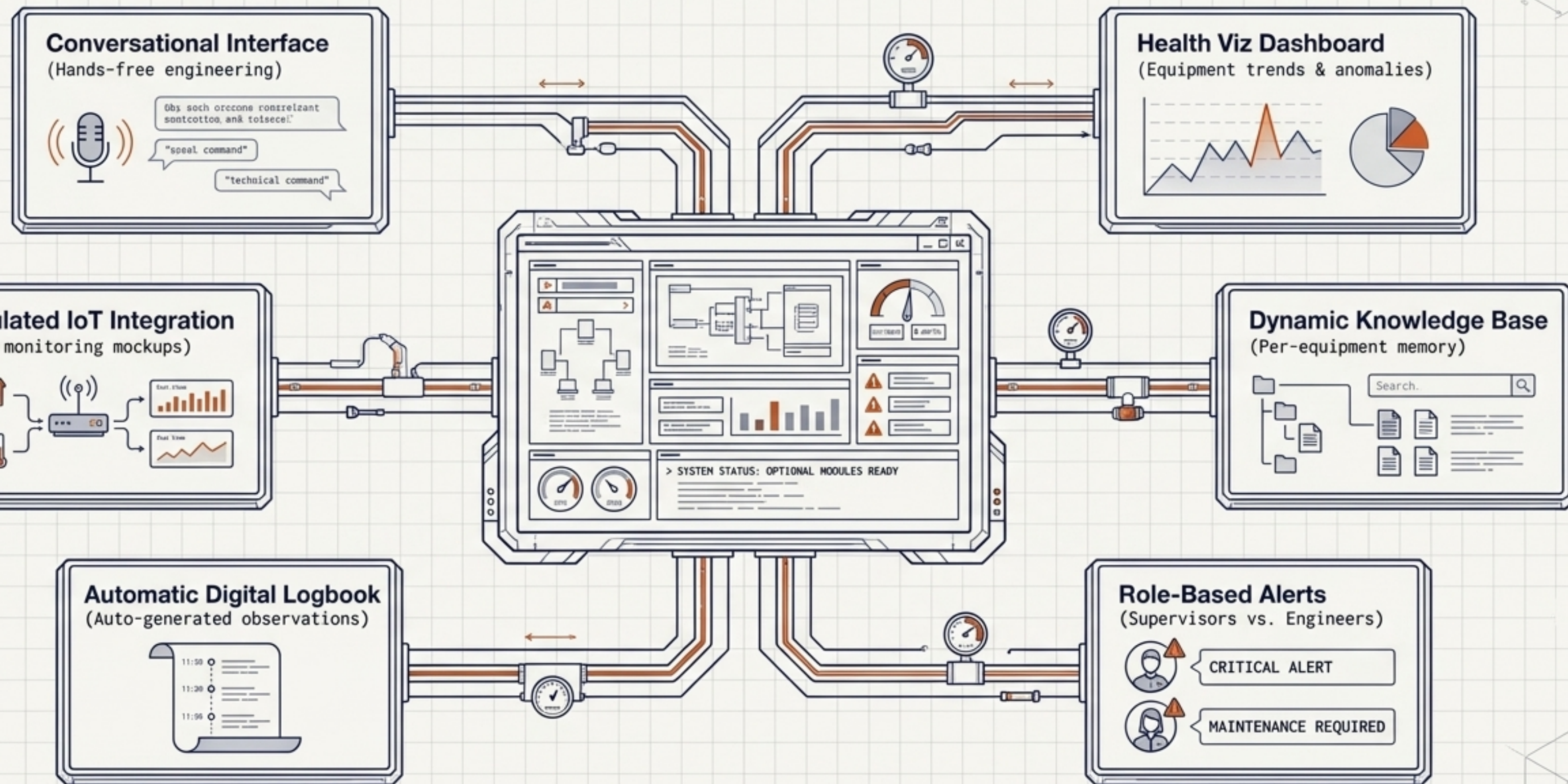
The Feedback Loop

User corrections, confirmations, and actual outcomes must be ingested back into the system to dynamically improve future recommendations.

The Continuous Improvement Flywheel

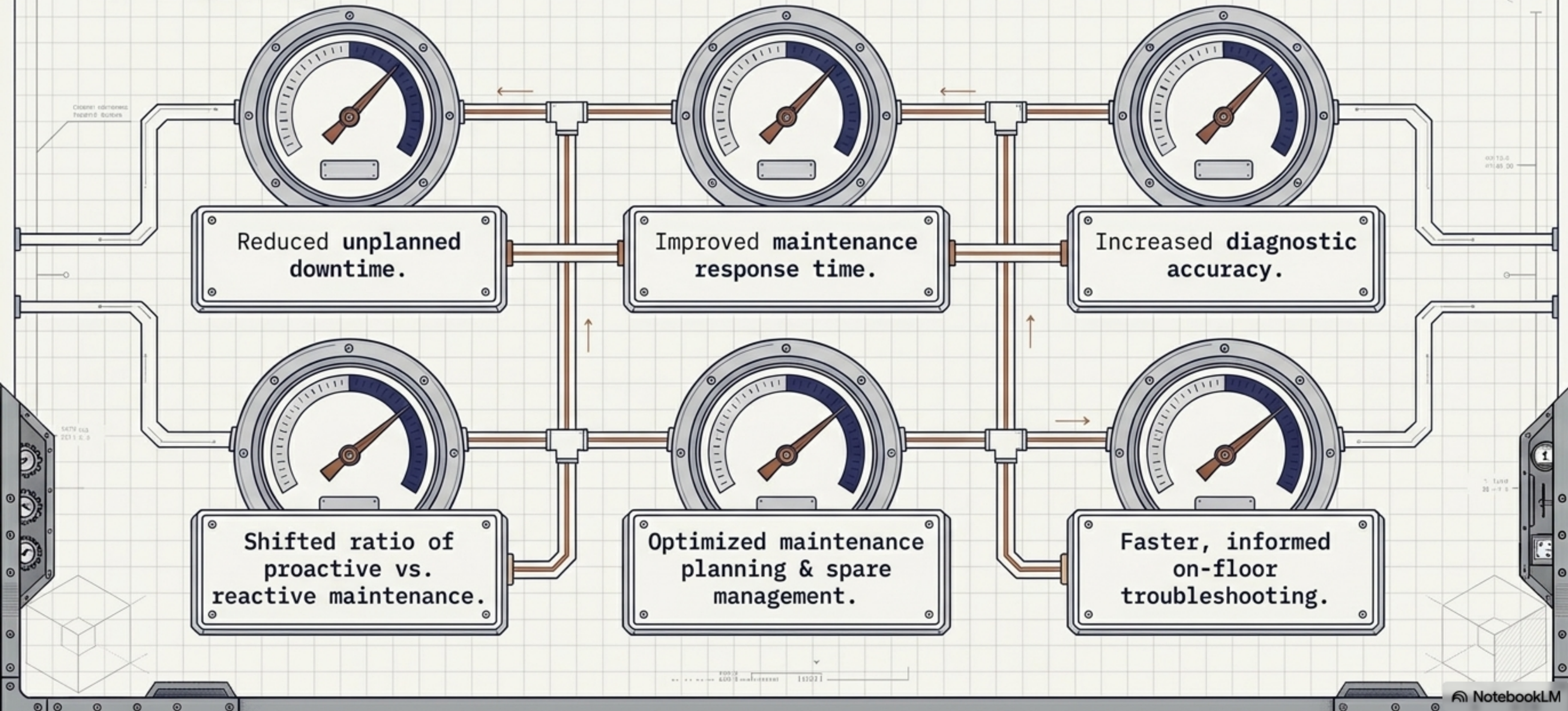


High-Impact Optional Enhancements



Teams may implement one or more of these modules to enhance the practical value of the system.

Expected Operational Outcomes



Deliverables Architecture Checklist

Submission.ZIP

Prototype_Code/

[Detailed source code required to run the working system.]

Demo_Recording.mp4

[Screen recording showcasing built features & sample I/O.]

Technical_Documentation.pdf

- System & Data Flow Architecture
- Technology Stack
- Model Design & Reasoning Pipeline
- Alerting & Prediction Logic
- Assumptions & Limitations
- Install & Config Instructions